

National Institute of Technology, Rourkela
End Semester Examination, Autumn 2024
Compiler Design (CS3007)

Duration: 3 Hours
Number of pages: 2

Full Marks: 50

- **Attempt any ten out of twelve questions.**
- **In a grammar, upper case letters are non-terminals, rest symbols are terminals.**

- Q 1. Find the deterministic finite automata directly from the regular expression $ba \circ (a|b)^* \circ ab\#$. (Using syntax tree and computation of $firstpos()$, $lastpos()$, and $followpos()$) [5]
- Q 2. Design a syntax-directed translation scheme check to whether a given string $w \in \{a^n b^n c^n\}$ where $n \geq 0$. Construct the annotated parse tree for $w = a^3 b^3 c^3$. [5]
- Q 3. Design a deterministic finite automata that will accept a valid date in $dd-mm-yyyy$ format where $dd \in \{01, 02, \dots, 31\}$, $mm \in \{01, 02, \dots, 12\}$ and $yyyy \in \{0000, \dots, 9999\}$. For example 30-02-2024 is invalid, and 29-02-2024 is valid. [5]
- Q 4. Apply loop-invariant code motion and strength reduction to the following code and rewrite it. [5]
- Q 5. Convert the code fragment of **Q 4** into three-address code assuming static storage allocation and the target machine is byte-addressed, with four bytes per word. [5]
- Q 6. Find the basic blocks and construct flow graph for the resultant three-address code in **Q 5**. Apply induction variable elimination, loop-invariant code motion and strength reduction to optimize the code. [5]
- Q 7. Write an intermediate code in the form of postfix notation for
if a + b then if c - d then c + d else x * y else if p then q + r else q * r [5]
- Q 8. Construct the DAG for the following basic block. Identify the common sub-expressions. [5]
D := B * C
E := A + B
B := B * C
A := E - D
- Q 9. Consider a flow graph $G(V, E)$ with n basic blocks as vertices $V = \{B_1, B_2, \dots, B_n\}$ and edges $E \subseteq V \times V$. Design an algorithm to find loops and unreachable codes in G . [5]
- Q 10. Construct an LL(1) parsing table for the language $\{a^n b^{2n} : n \geq 0\}$. Identify the error entries in the parsing table to deal with erroneous input strings such as $aabbbb$ and $aaabbbb$. [5]
- Q 11. Construct an SLR(1) parsing table for the language $\{a^n b^n : n \geq 0\}$. Identify the error entries in the parsing table to deal with erroneous input strings such as $aabbbb$ and $aaabb$. [5]
- Q 12. A machine μ has four general purpose register bank $R = \{r_0, r_1, r_2, r_3\}$ and a set of memory variables $M = \{m_0, m_1, \dots\}$. The instruction set comprises of following instructions
CMP X, Y ; compare X and Y where $(X \in R) \wedge (Y \in (R \cup M))$
JGT Ln ; jump/branch if-greater-than to label Ln where $n \in \{1, 2, \dots\}$
JMP Ln ; unconditional jump/branch to label Ln where $n \in \{1, 2, \dots\}$
SUB X, Y ; $X \leftarrow X - Y$ where $(X \in R) \wedge (Y \in (R \cup M))$

MOV $X, Y; X \leftarrow Y$ where $((X \in R) \wedge (Y \in (R \cup M))) \vee ((X \in (R \cup M)) \wedge (Y \in R))$

Generate target code for μ from the three-address code of $c = (a > b)?a - b : b - a$; using labeled tree method, where a, b and c are memory variables in M . [5]

*******All the best*******